New Attacks on QR-UOV Across Vinegar Dimensions: Direct Forgery and Equivalent-Key Recovery

Justin Thaler* Kai Zhang† Scott Kominers‡ Quang Dao§

August 7, 2026

Abstract

QR-UOV is a structured variant of Unbalanced Oil and Vinegar (UOV) in the third round of NIST's Additional Digital Signatures process. It retains UOV's short signatures and simple signing procedure while compressing its large public key through blocks over a degree-three extension field. Its security must account for two attacks. A direct forger solves the public equations for one valid signature. A key-recovery attacker reconstructs a hidden oil space that permits signatures on every message. The vinegar count affects these attacks in opposite directions. Increasing it makes the hidden oil space harder to recover, but gives a direct forger more variables with which to solve the public equations.

In this paper, we analyze the two attacks together and prove one bound that holds for every vinegar count. Let m be the number of public equations and ℓ the extension degree. For every vinegar count and every nonzero verification target, one of the two reductions produces a square system with at most $m - \ell - 1$ equations and variables. Changing the vinegar count can move the smaller system from one attack to the other, but cannot raise this bound. In QR-UOV, $\ell = 3$, so the bound is $m - 4$, or 50, 74, and 101 variables for the three recommended parameter sets. This bounds the system dimension, not total running time or memory.

We estimate concrete attack costs under the Boolean-gate model used in the QR-UOV specification. The fastest estimates fall below the three claimed category targets by 2.6, 16.0, and 24.9 bits. These estimates are conditional and require more memory than any physical system can provide, so they do not constitute practical full-parameter attacks. The variable reduction is independent of the solver. A low-memory matrix solver uses between 2.4 and 9.5 MiB, but its running times remain prohibitive. It still improves the previous low-memory time estimates by 17 to 31 bits. On smaller parameters, our implementation produces forged signatures that the unchanged verifier accepts, confirming that the algebraic steps work together.

1 Introduction

NIST opened its Additional Digital Signatures process to broaden the set of post-quantum signatures beyond the structured-lattice schemes selected in its first standardization process. After the second round, NIST advanced nine candidates, including QR-UOV, to a third round of public analysis [11, 12]. QR-UOV retains the short signatures and simple signing procedure of Unbalanced Oil and Vinegar (UOV) while reducing UOV's main efficiency cost: its large public key [1, 3].

UOV divides the secret coordinates into vinegar variables and oil variables. Its central quadratic equations contain no product of two oil variables. After choosing the vinegar variables, the signer

*Georgetown University and a16z crypto research.
†MIT.
‡Harvard University and a16z crypto research.
§Carnegie Mellon University and LayerZero Labs.

can therefore solve a linear system for the oil variables. A secret change of coordinates hides this division in the public equations. Using more vinegar variables can make the hidden oil space harder to recover, but it also gives a direct forger more variables with which to solve the public equations [4, 5, 6].

QR-UOV compresses the UOV public key by grouping base-field coordinates into blocks over a degree-ℓ extension field. Let V be the number of extension-field vinegar blocks and let m be the number of public quadratic equations over the base field. The current QR-UOV construction uses $\ell = 3$. This representation reduces the amount of public-key material, but the verifier still evaluates the same kind of dense quadratic map as in UOV [1, 3].

This structure gives an attacker two natural goals. A direct forger needs only one input that maps to a chosen verification target. A key-recovery attacker instead seeks any hidden oil space that reproduces the public map and therefore permits signatures on every target. These goals respond oppositely to the vinegar dimension: increasing V can make the hidden oil space harder to recover, but it also leaves more variables available to a direct forger. The exact QR-UOV equations and signing interface appear in Section 3.

The public key still represents m dense quadratic forms, each with quadratically many coefficients. Field lifting and block-anti-circulant UOV are earlier approaches to compressing these coefficients [7, 8]; the latter was broken by decomposing the reducible quotient ring behind its public blocks [9]. QR-UOV prevents that decomposition by using an irreducible polynomial, but the public map can still be viewed over the extension field. This view underlies attacks that translate the map between the base and extension fields, as well as attacks that search a public linear space for low-rank matrices [1, 17].

Existing work studies both consequences of changing V, but mostly in separate attack families. Direct attacks use the total number of variables: as V grows, the public equations become more underdetermined and can be reduced to a smaller residual system [20, 27, 28, 29]. Structural attacks use the hidden oil space: as V grows, finding enough oil vectors generally becomes harder [5, 10, 16, 23]. Pébereau states this tradeoff explicitly and proves that, in the usual parameter range, one suitable oil vector determines an equivalent UOV key [23]. The original QR-UOV analysis likewise chooses the vinegar count mainly against structural attacks and the output count against direct forgery [1].

These are close precedents for the paper's motivation. They do not, however, join the two sides in one statement quantified over every integer V. The specification and the team's latest security response instead evaluate the attacks at fixed recommended parameter sets [3, 26]. We ask what can be guaranteed when V itself varies while m and ℓ remain fixed.

We call a nonlinear system *square* when it has the same number of equations and variables. We compare the two attack families by counting the variables in the square system that each attack must solve. This brings us to the question that governs the paper:

> **Question.** *Can QR-UOV force both attack reductions to use square nonlinear systems with more than $m - \ell - 1$ variables by changing only the vinegar count, while keeping the number of public equations m and the extension degree ℓ fixed?*

1.1 Our results and technical overview

We answer this question negatively. Our main result is the following bound:

For every vinegar count and nonzero verification target, either the reduction for direct forgery or the reduction for equivalent-key recovery produces a square nonlinear system with at most $m - \ell - 1$ variables.

For QR-UOV, $\ell = 3$, so the bound is $m - 4$ variables. The statement, formalized in Theorem 4.10, concerns the shared nonlinear system. It does not assert that total running time or memory depends only on this dimension.

Direct forgery. For a fixed nonzero target, we first change the public output coordinates so that all but one target coordinate are zero. We then use Chevalley–Warning to construct an $(m-3)$-dimensional subspace on which three of the corresponding quadratic forms vanish. The remaining zero-target equations are homogeneous, so multiplying a solution by a nonzero scalar does not change them. Fixing this common scale leaves the same number of equations and variables, namely $m - 4$. The dimensions are 50, 74, and 101 for the three recommended parameter sets (Theorem 4.9 and Remark 4.1). The attack uses only public randomness, and the verifier checks every returned vector against the original public map (Section 4).

Equivalent-key recovery. In extension-field coordinates, the hidden oil space is the graph of a linear map G. Fixing a public oil vector c produces m public quadratic equations in V variables. The secret graph guarantees that Gc is a root, although the attacker does not know this root in advance. If the matrix of first derivatives has full rank at a recovered root, public linear equations determine all of G. The pull-back and signing checks then certify another secret key that signs every target accepted by the verifier (Theorem 7.3). Pébereau established the equivalent-key endpoint from one suitable oil vector [23]; our contribution is a public polynomial-system mechanism for finding and certifying the required information in QR-UOV.

Why vinegar-only retuning cannot help. The direct-forgery and key-recovery routes respond oppositely to V. When $V \le m - \ell - 1$, key recovery uses at most $m - \ell - 1$ variables. When $V \ge m - \ell$, the direct reduction and fixing the common scale use exactly $m - \ell - 1$. These cases cover every integer V, so

$$r_{\text{attack}}(V) \le m - \ell - 1.$$

For QR-UOV, $\ell = 3$, so the ceiling is $m - 4$ (Theorem 4.10). Increasing only V therefore cannot move the two attacks past their crossing point. A parameter revision must also increase m, or change the construction so that one of the reductions no longer applies.

Concrete estimates and executable evidence. The reductions above still leave large nonlinear systems. To estimate the work required to solve them, we adapt an exact symbolic equation solver to the base field of characteristic 127. The solver starts from equations whose roots are known, changes them gradually into the attack equations, and tracks the roots through this change. We also account for its polynomial arithmetic over extension fields. Under the specification's Boolean-gate cost convention, the resulting estimates are $\mathbf{2^{140.384}}$, $2^{191.027}$, and $2^{247.125}$ gates at Levels I, III, and V, respectively. These rows lie 2.616, 15.973, and 24.875 bits below their category targets (Table 3). The Level-I row is a candidate break only within this cost model. It depends on extrapolating a degree bound to the exact attack equations and on an unimplemented fast polynomial multiplication schedule. The symbolic representation also needs an impractical amount of memory. Because the reduction is independent of the solver, we also apply polynomial XL (PXL) and Wiedemann XL (WXL), two standard methods for solving quadratic systems. PXL gives $143.643, 204.845, 272.975$ bit estimates. A low-memory WXL setting first guesses many variables and then uses sparse linear algebra. It uses under 10 MiB at every level and improves the running time of the previous low-memory route (Sections 8 and 10).

On smaller examples, the direct-forgery implementation runs from public input to a forged signature accepted by the verifier over two small fields. This checks that the attack steps work together, but it does not validate the full-parameter cost estimate (Section 9). The algebraic reductions and final verification are exact. The success probabilities use the stated random-key models, and the gate counts remain model-based estimates.

Organization. Section 2 positions the result against prior attacks, and Section 3 gives the exact QR-UOV notation and attack interface. Sections 4 and 5 prove the two reductions and the fixed-output bound. The remaining sections give the root solver, concrete estimates, evidence, limitations, and parameter consequences.

2 Related work

From Oil and Vinegar to QR-UOV. UOV was introduced after the balanced construction's hidden linear structure was recovered; its vinegar imbalance must be chosen against both structural attacks and an increasingly underdetermined public system [4, 5, 6]. Field lifting and block-anti-circulant matrices compress the public key [7, 8], but the latter construction was broken through its reducible quotient polynomial [9]. QR-UOV instead uses an irreducible quotient and analyzes the resulting pull-back and lifting attacks [1]. We hold this construction fixed and vary only its vinegar dimension.

Direct attacks on underdetermined quadratic systems. Thomae and Wolf reduce an underdetermined quadratic system to a smaller square system before invoking a generic solver [28]. Write $\omega = n/m$. Their main reduction leaves $m - \lfloor \omega \rfloor + 1$ equations and variables; when $\lfloor \omega \rfloor$ divides m, their tighter analysis leaves $m - \lfloor \omega \rfloor$. They prove this over even characteristic and outline an odd-characteristic adaptation for small ω. Thus the work already measures how extra variables make direct forgery easier, though only through the coarse ratio n/m. Furue, Nakamura, and Takagi combine the reduction with guessed variables, and Hashimoto partitions the residual equations into smaller systems over general finite fields [27, 29]. These works optimize a direct attack for one fixed pair (n, m); they do not compare that attack with key recovery as the vinegar count varies.

For the present $q = 127$ parameter sets, the Round-2 specification estimates generic equation solving with Wiedemann XL (WXL), which uses sparse linear algebra, and polynomial XL (PXL), which first eliminates over a polynomial ring. It reports classical costs of $160, 222, 288$ bits for Hashimoto with WXL and $150, 211, 279$ bits for PXL [3]. The team's January 2026 response updates the Hashimoto row to $158, 217, 285$ bits [26]. The affine residual dimensions $51, 75, 102$ are therefore prior art. The later "Just Guess" algorithm has classical costs $258, 399, 561$ bits on the same $q = 127$ systems; its main gain there is quantum [20, 21]. Our direct attack starts from the prior affine dimensions. We make the residual equations homogeneous after changing the output coordinates, so solutions are defined only up to a common nonzero scale. Fixing that scale removes one further variable and gives systems in $50, 74, 101$ variables.

Time and memory of the direct attacks. The published numbers above are time estimates, not complete time and memory points. We reconstruct the missing storage from the matrix dimensions in the published formulas. Even when entries are regenerated, the Level-I Hashimoto matrix needs about 29 PiB for three vectors, while the usual PXL state needs about 4.8 ZiB. Just Guess is different. It walks its search tree depth first and needs only the quadratic coefficients, its change of basis, and one search path. This is on the order of tens of MiB [20].

Our fastest solver stores much more because it follows every algebraic branch at once. The exact reduction does not require this choice. Applying PXL to our $50, 74, 101$ variable systems saves about six time bits and about four storage bits over the specification's PXL instances. Applying WXL and regenerating its matrix entries yields estimates of $238.387, 367.819, 543.595$ bits with $2.4, 6.9, 9.5$ MiB. These rows are respectively $19.613, 31.181, 17.405$ bits faster than Just Guess. Thus the reduction shifts both the fast end and the low-memory end of the modeled frontier (Tables 4 and 5).

Recovering the hidden oil space. The Kipnis–Shamir attack searches directly for the secret linear structure [5]. Reconciliation expresses membership in the oil space as quadratic equations. Beullens' intersection attack instead compares several transformed copies of the oil space and uses their intersections to reduce the search [10, 16]. Pébereau makes the dependence on the vinegar dimension precise. Writing $n = v + m$, the Kipnis–Shamir route costs polynomial time at $n = 2m$ and about $q^{n-2m}n^{\omega} = q^{v-m}n^{\omega}$ when $n > 2m$. One recovered oil vector determines an equivalent key in $O(mn^{\omega})$ operations when $n \leq 3m$; for $n > 3m$, more oil vectors are needed [23]. Beullens' basic intersection attack likewise applies in the regime $n < 3m$, or $v < 2m$, with stronger multiway variants in narrower ranges [16]. These results show explicitly that structural recovery becomes harder as v grows. We do not claim this tradeoff or the one-vector endpoint. Our contribution is a public QR-UOV mechanism for finding and certifying the required information. A chosen extension-field direction exposes a planted root Gc, the derivative test certifies when that root determines the hidden graph, and exact signing checks certify the recovered equivalent key (Section 5).

QR-UOV-specific key recovery. The original QR-UOV analysis applies reconciliation and intersection after pulling the block matrices back to the quotient field or lifting the public map to an extension field [1]. Furue and Ikematsu instead encode the same structure as the problem of finding a low-rank matrix in a public linear space [17]. Suzuki et al. extend the admissible target ranks and connect this method to the singular points used by other UOV attacks [18]. More recent methods derive higher-degree algebraic relations using symmetric powers or truncated polynomial rings. These relations improve key recovery for some UOV parameter sets [19, 32]. For the recommended QR-UOV sets, the truncated-ring calculation gives no improvement because the characteristic 127 already exceeds the relevant solving degree, and the reported odd-characteristic symmetric-algebra costs do not beat reconciliation. Ran's wedge attack gives one of the few explicit memory analyses: if its sparse $N \times N$ matrix has ρ nonzero entries per row, storing the matrix costs $O(\rho N(\log N + \log q))$ bits, while regenerating entries reduces working memory to $O(N \log q)$ bits at additional non-field overhead [22]. For QR-UOV it costs 2^{182}, 2^{249}, and 2^{328} field operations and does not improve the security estimates. These works sharpen individual key-recovery costs at fixed parameters. They use the vinegar count inside their matrix dimensions or applicability conditions, but do not optimize over every integer V or join key recovery to direct forgery.

Parameter tradeoffs and common UOV frameworks. Pébereau and the original QR-UOV parameter analysis already state the qualitative tension between key recovery and forgery [1, 23]. Li, Guo, and Ding describe QR-UOV, MAYO, and SNOVA in one algebraic framework and show that QR-UOV key-recovery costs can be analyzed as UOV over the extension field [2]. We claim neither the tradeoff nor the extension-field view. Our result is the exact two-case bound: every integer V leaves either the direct or equivalent-key reduction with at most $m - \ell - 1$ variables, and the direct reduction reaches this bound only after fixing the common scale (Theorem 4.10).

Polynomial-system solving. We estimate root-finding costs with the exact symbolic algorithm of Safey El Din and Schost [24]. It connects an easy system with known roots to the target system and follows those roots as the connecting parameter changes [14, 25]. We change the easy starting system and the linear form used to distinguish roots so that the algorithm works in characteristic 127. The supplement checks the result against two other exact solvers [13, 15]. These solvers estimate the cost of the systems produced by our attacks. They do not produce either QR-UOV reduction or the fixed-output theorem.

Chevalley–Warning. The direct reduction uses the classical fact that, when the sum of the degrees is smaller than the number of variables, the number of common zeros over a finite field is divisible by the characteristic [30, 31]. Applied to three quadrics in seven variables, it guarantees a nonzero extension vector. Exhaustive search over nonzero vectors up to scalar multiplication makes the theorem constructive, so the preprocessing does not assume that a common zero is available for free. Our use of this classical theorem is therefore algorithmic rather than a new finite-field existence result.

3 QR-UOV and the attack interface

This section gives the exact QR-UOV notation used by both attacks. We first define the secret and public quadratic maps and the signing equation. We then state the random expanded-key models used only for success probabilities and give the public checks that turn a recovered oil space into a signing key.

3.1 Parameters, public map, and signing

Fix an odd prime power q and an irreducible polynomial $f \in \mathbb{F}_q[X]$ of degree ℓ. Write

$$K := \mathbb{F}_q[X]/(f) \cong \mathbb{F}_{q^\ell}, \qquad v = \ell V, \qquad m = \ell O, \qquad N = V + O, \qquad n = v + m = \ell N. \tag{1}$$

The verifier evaluates m quadratic forms in n variables over $\mathbb{F}_q$. The quotient representation groups these variables into V vinegar blocks and O oil blocks over K. The number of output forms remains m, rather than falling to $O = m/\ell$.

For each output $i \in [m]$, the secret central matrix and the secret input transformation have the following form over K:

$$F_i = \begin{pmatrix} A_i & B_i \\ B_i^\top & 0 \end{pmatrix}, \qquad S_G = \begin{pmatrix} I_V & G \\ 0 & I_O \end{pmatrix}, \qquad P_i = S_G^\top F_i S_G = \begin{pmatrix} A_i & E_i \\ E_i^\top & C_i \end{pmatrix}. \tag{2}$$

Here $G \in K^{V \times O}$ is secret, $E_i = A_i G + B_i$, and block multiplication gives

$$C_i = G^\top E_i + E_i^\top G - G^\top A_i G. \tag{3}$$

The zero lower-right block of F_i is the UOV condition. For extension-field vinegar and oil coordinates $(z, c) \in K^V \times K^O$, it gives

$$F_i(z, c) = z^\top A_i z + 2 z^\top B_i c. \tag{4}$$

Once z is fixed, every central quadratic form is linear in c.

The same matrices identify the hidden oil space in the public coordinates. It is the graph

$$\mathcal{O} = S_G^{-1}(\{0\}^V \times K^O) = \left\{ \begin{pmatrix} -Gc \\ c \end{pmatrix} : c \in K^O \right\}. \tag{5}$$

Every public form vanishes on every pair of vectors from this space. In matrix notation, this means

$$u^{\mathsf{T}} P_i w = 0 \qquad (u, w \in \mathcal{O},\ 1 \le i \le m). \tag{6}$$

Conversely, a matrix G' defines a common oil graph for the public tuple exactly when it satisfies (3) with G' in place of G. Thus (3) is the condition that any recovered oil graph must pass.

We now convert the extension-field matrices into the base-field matrices used by the signer and verifier. The regular representation $\Phi_f : K \to \mathbb{F}_q^{\ell \times \ell}$ maps an element $g \in K$ to the matrix for multiplication by g on its ℓ base-field coefficients. The map $\Phi_f^{(a,b)}$ applies this conversion to every block of a matrix in $K^{a \times b}$. The specification also supplies a matrix W_f that makes $W_f \Phi_f(g)$ symmetric for every $g \in K$. Let $W_f^{(a)}$ contain a copies of W_f on its diagonal. The specification's conversion, which we call the *pull-back*, gives the following base-field matrices:

$$\widehat{P}_i = W_f^{(N)} \Phi_f^{(N,N)}(P_i), \qquad \widehat{F}_i = W_f^{(N)} \Phi_f^{(N,N)}(F_i), \qquad \widehat{S}_G = \Phi_f^{(N,N)}(S_G). \tag{7}$$

The pull-back preserves the secret factorization, so

$$\begin{aligned} \widehat{P}_i &= \widehat{S}_G^{\mathsf{T}} \widehat{F}_i \widehat{S}_G, \\ \widehat{P}(x) &:= (x^{\mathsf{T}} \widehat{P}_1 x, \dots, x^{\mathsf{T}} \widehat{P}_m x), \\ \widehat{F}(z) &:= (z^{\mathsf{T}} \widehat{F}_1 z, \dots, z^{\mathsf{T}} \widehat{F}_m z), \qquad \widehat{P}(x) = \widehat{F}(\widehat{S}_G x). \end{aligned} \tag{8}$$

A general $\ell \times \ell$ base-field block has ℓ^2 entries, while $\Phi_f(g)$ is determined by the ℓ coefficients of $g \in K$. The official public seed expands the blocks A_i and E_i, the public key stores the remaining blocks C_i, and the secret seed expands G. This is how QR-UOV compresses the dense UOV verification map without changing the map that the verifier evaluates [1, 3].

The signing equation will also be the interface for both attacks. For a message M and salt r, define the verification target

$$\mu = \mathrm{SHAKE256}(\mathsf{seed}_{\mathrm{pk}} \parallel \mathrm{BytesToBits}(M), 512), \qquad y(M, r) = \mathrm{Hash}(\mu, r) \in \mathbb{F}_q^m. \tag{9}$$

For a vinegar assignment $a \in \mathbb{F}_q^v$, define the oil coefficient matrix by

$$\mathsf{L}_{\widehat{F}}(a) \in \mathbb{F}_q^{m \times m}, \qquad \left(\mathsf{L}_{\widehat{F}}(a) o\right)_i := \widehat{F}_i(a, o) - \widehat{F}_i(a, 0). \tag{10}$$

The right-hand side is linear in $o \in \mathbb{F}_q^m$ because the oil block of $\widehat{F}_i$ is zero. The signer chooses a and then samples r until

$$\mathsf{L}_{\widehat{F}}(a) o = y(M, r) - \widehat{F}(a, 0) \tag{11}$$

has a solution [3, Sec. 4.2]. It returns the signature (r, x) with $x = \widehat{S}_G^{-1}(a, o)$. The verifier's acceptance condition is [3, Sec. 4.3]

$$\widehat{P}(x) = y(M, r). \tag{12}$$

The verifier equation exposes the two attack goals studied here. Direct forgery fixes a nonzero target y and finds one x satisfying $\widehat{P}(x) = y$. Equivalent-key recovery finds any G' satisfying (3), pulls the resulting central forms back to $\mathbb{F}_q$, and finds an a for which $\mathsf{L}_{\widehat{F}}(a)$ is invertible. The same linear system then signs every verification target.

3.2 Recommended parameters and expansion models

The Round-2 parameter sets specialize (1) to $q = 127$ and $\ell = 3$. Thus

$$K = \mathbb{F}_{q^\ell} = \mathbb{F}_{127^3}, \qquad Q = |K| = 2{,}048{,}383, \qquad \log_2 Q = 20.966054\ldots.$$

The three parameter sets are as follows.

Level	v	m	V	O	N	$e = m - V$
I	156	54	52	18	70	2
III	228	78	76	26	102	2
V	306	105	102	35	137	3

We write $D = 2^V$ for Bézout's general upper bound on the number of isolated roots of V quadratic equations in V variables. We write $L_s = \mathbb{F}_{Q^s}$ for an extension field used by the equation solver. All concrete theorems below concern these three sets, for which $m < q$. The matrices A_i, B_i, E_i, C_i, the graph G, and the oil space $\mathcal{O}$ retain the meanings in (2) to (6).

For the probability calculations, we replace the seeded expansion by the following explicit distribution on expanded keys. The algebraic attacks and all final verification checks do not depend on this replacement.

Definition 3.1 (Random expanded-key model)**.** Sample the symmetric blocks $A_1, \dots, A_m$ independently and uniformly from $\mathrm{Sym}_V(K)$. Conditional on $A = (A_i)_i$, sample the secret graph $G \in K^{V\times O}$ from any distribution. Conditional on (A, G), sample $E_1, \dots, E_m$ independently and uniformly from $K^{V\times O}$, and set each C_i by (3). No independence between G and A is assumed.

Definition 3.2 (Independent-graph model)**.** The independent-graph model is Definition 3.1 with G independent of A. The marginal distribution of G may otherwise be arbitrary.

Proposition 3.3 (Central blocks remain uniform)**.** In Definition 3.1, the blocks $B_i = E_i - A_i G$ are independent and uniform in $K^{V\times O}$, and $B = (B_i)_i$ is independent of (A, G). Hence the pairs $(A_i, B_i)_{1\le i\le m}$ are independent and uniform on $\mathrm{Sym}_V(K) \times K^{V\times O}$, although G may remain correlated with A.

Proof. For fixed (A, G), each map $E_i \mapsto E_i - A_i G$ is a translation of $K^{V\times O}$. It therefore preserves the uniform law and makes that law independent of (A, G). Independence of the A_i and E_i gives the stated product distribution. □

The equivalent-key analysis uses the first model. Its failure events depend on the uniform central matrices and on fresh public attack randomness, but not on independence between G and A. The direct reduction is valid for every key. Its average success probability, however, requires G to be independent of the central coefficients and therefore uses the second model. The supplement explains when idealized random functions or ideal ciphers justify these distributions and why the argument does not automatically transfer to the fixed SHAKE or AES functions (Section G). These models affect only the probability of finding a candidate. Every returned key and signature is checked against the original public map.

3.3 From a recovered graph to a signing key

The graph identities are useful because they can be checked using only public matrices. Passing these checks reconstructs a central UOV map for the same public key.

Proposition 3.4 (A verified graph gives an exact UOV decomposition)**.** Let $G' \in K^{V\times O}$ satisfy every identity (3), and put

$$B'_i = E_i - A_i G', \qquad F'_i = \begin{pmatrix} A_i & B'_i \\ B_i'^{\mathsf{T}} & 0 \end{pmatrix}, \qquad S_{G'} = \begin{pmatrix} I_V & G' \\ 0 & I_O \end{pmatrix}.$$

Then $P_i = S_{G'}^{\mathsf{T}} F'_i S_{G'}$ for every i. The specification's invertible base-field conversion therefore gives matrices $\widehat{P}_i = \widehat{S}_{G'}^{\mathsf{T}} \widehat{F}'_i \widehat{S}_{G'}$ over $\mathbb{F}_q$ for the unchanged verification map [3, Sec. 4.1].

Proof. Direct block multiplication gives the extension-field factorization because the lower-right block is zero exactly when (3) holds. Applying the specification's inverse pull-back gives the base-field factorization. The full matrix calculation appears in Section F. □

Proposition 3.5 (Public signing witness)**.** If $\det \mathsf{L}_{\widehat{F}}(a) \neq 0$, then $(\widehat{F}, \widehat{S}, a)$ is a publicly checkable signing certificate. For every target $y \in \mathbb{F}_q^m$, the unique solution to

$$o = \mathsf{L}_{\widehat{F}}(a)^{-1}(y - \widehat{F}(a, 0)), \qquad x = \widehat{S}^{-1}(a, o) \tag{13}$$

satisfies $\widehat{P}(x) = y$.

Proof. Substitution gives $\widehat{F}(a, o) = y$ and then $\widehat{P}(x) = \widehat{F}(\widehat{S}x) = y$. The forged signatures need not match the honest signing distribution. □

Corollary 3.6 (Forging any chosen message without signing queries)**.** A verified certificate from Proposition 3.5 lets an attacker forge an accepted QR-UOV signature on every chosen message without a signing query. Given a message M, choose any allowed salt r, form $y(M, r)$ by (9), invert it with Proposition 3.5, and output (r, x).

Proof. The certificate satisfies the verifier equation (12) for every target. It neither recovers nor uses the secret seed. □

3.4 Finding vinegar values with an invertible oil system

Fix nonzero $\xi \in \mathbb{F}_q^\ell$. For each quotient-ring vinegar block j, let a_j place ξ in block j and zero elsewhere, and set

$$M_j := \mathsf{L}_{\widehat{F}}(a_j) \in \mathbb{F}_q^{m \times m}.$$

Write

$$\pi_m := \prod_{k=1}^{m}(1 - q^{-k}), \qquad \sigma_m := 1 - \pi_m.$$

π_m is the probability that a uniform $m \times m$ matrix over $\mathbb{F}_q$ is invertible. We first test vinegar assignments supported on one quotient-ring block. This makes the search both public and inexpensive.

Proposition 3.7 (Searching basis vinegar assignments)**.** For the planted decomposition in Definition 3.1, the matrices $M_1, \ldots, M_V$ are independent and uniform. The following searches then give public signing certificates.

1. Testing basis assignments in order uses fewer than $\pi_m^{-1} < 1.008$ tests on average. Testing all V fails with probability σ_m^V, whose base-two logarithms at Levels I, III, and V are -362.823, -530.280, and -711.692.

2. Testing only the first 22, 31, and 40 assignments at the three levels fails with base-two log probabilities
$$-153.502, \qquad -216.298, \qquad -279.095. \tag{14}$$

3. Pairing the basis matrices as $(M_1, M_2), (M_3, M_4), \ldots$ gives a deterministic fallback. Let $\mathcal{E}_{\text{sign}}$ be the event that every chosen pair span contains only singular matrices. Its probability satisfies
$$\Pr[\mathcal{E}_{\text{sign}}] \leq \eta_{\text{sign}}. \tag{15}$$
Its base-two logarithms at Levels I, III, and V are below
$$-544.820, \qquad -796.276, \qquad -1068.687.$$

Outside this event, testing every M_j and then $A + tB$ for each chosen pair and $t \in \mathbb{F}_q^\times$ succeeds after at most $V + \lfloor V/2 \rfloor (q-1)$ tests.

For any exact candidate, define

$$\Delta(c_1, \ldots, c_V) := \det\left(\sum_{j=1}^{V} c_j M_j\right).$$

The coefficients $(c_1, \ldots, c_V)$ correspond to the vinegar assignment $a = (c_1\xi, \ldots, c_V\xi)$. If Δ is nonzero, a uniform assignment of this form succeeds with probability at least $1 - m/q$. The expected numbers of assignments are at most 1.740, 2.592, and 5.773 at the three levels. This last claim is key by key, and every successful determinant test is an exact public certificate even when the recovered graph differs from the original secret graph.

Proof. The specification's regular representation maps each independent uniform entry of the central block B_i bijectively to one row block of M_j. Thus the M_j are independent and uniform. The remaining statements follow from the exact rank distribution of a uniform matrix, a count of singular two-dimensional spans, and Schwartz–Zippel. The formulas and calculations appear in Section F. □

4 Direct forgery in $m - 4$ variables

The direct attack starts from the verifier equation (12) over $\mathbb{F}_q$, with $q = 127$ and $n = v + m$. Given a nonzero target y, it constructs an $(m-3)$-dimensional public subspace on which three zero-target equations vanish identically. On this subspace, the remaining zero-target equations are homogeneous. Their solutions are therefore defined up to multiplication by a nonzero scalar. Fixing this scale leaves $m - 4$ equations in $m - 4$ variables. The final nonzero-target equation recovers the omitted scale. The resulting systems have dimensions

$$50, \qquad 74, \qquad 101$$

for Levels I, III, and V.

Remark 4.1 (Relation to Hashimoto and the specification)*.* The affine reduction to $m-3$ variables is prior art. Hashimoto's underdetermined quadratic-system method and the QR-UOV specification already analyze these residual dimensions [3, 27]. Our additional reduction uses full target normalization and solves the homogeneous zero-target equations modulo scaling. This removes one more variable, giving systems of dimension $m - 4$. We also give a finite construction of the required subspace, bound its failure probability for every fixed key, prove a lower bound on the probability of a root with invertible Jacobian, and estimate the cost over the base field of characteristic 127.

The algebraic reduction below is key by key. Its probability calculation uses the independent-graph model of Definition 3.2: the hidden graph G is independent of the independent uniform central coefficient pairs (A_i, B_i). This independence is needed only for the direct attack's average success analysis. The equivalent-key results continue to allow arbitrary correlation between G and A. In idealized models where SHAKE is a random function or AES is an ideal cipher, independently sampled and distinct public and secret seeds give the required independence. Section G accounts for seed collisions and finite output lengths. We do not claim that fixed SHAKE or AES automatically has the same distribution.

4.1 From the target to a projective system

If $y = 0$, then $x = 0$ is already a valid preimage, so only the nonzero case needs analysis. Let $y \in \mathbb{F}_q^m$ be nonzero. Choose a public matrix $T_y \in \mathrm{GL}_m(\mathbb{F}_q)$ whose first $m-1$ rows span $y^\perp$ and whose last row has inner product one with y. Then

$$T_y y = (0, \dots, 0, 1)^\mathsf{T}. \tag{16}$$

Write $Q_1, \dots, Q_m$ for the corresponding output recombinations of the public forms. Since T_y is fixed by y and invertible, independent uniform public forms remain independent and uniform in the random expanded-key model.

Only Q_1, Q_2, Q_3 are used to construct the attack subspace. Choose a uniformly random seven-dimensional subspace $S_0 \subset \mathbb{F}_q^n$ and restrict these three forms to S_0. The attacker then tests all nonzero vectors in S_0 up to scalar multiplication, namely the points of $\mathbb{P}(S_0) \simeq \mathbb{P}^6(\mathbb{F}_q)$. Chevalley–Warning guarantees a nonzero common zero because three quadrics have total degree six in seven variables. Denote the first such point by w.

The hidden oil space is not known to the attacker, so the algorithm cannot test whether w lies outside it. We call such a point *off oil.* The random seed subspace makes this test unnecessary; the next proposition bounds the failure probability for every fixed oil space.

Proposition 4.2 (A random seed misses every fixed oil space)**.** Fix an arbitrary m-dimensional subspace $\widehat{\mathcal{O}} \subset \mathbb{F}_q^n$ and choose a uniform seven-dimensional subspace $S_0 \subset \mathbb{F}_q^n$. Then

$$\Pr[S_0 \cap \widehat{\mathcal{O}} \neq \{0\}] \leq \epsilon_{\text{seed}} := \frac{q^m - 1}{q - 1} \, \frac{q^7 - 1}{q^n - 1}. \tag{17}$$

Consequently, except with probability at most ϵ_{seed} over the attack's public randomness, every nonzero common zero found inside S_0 is off oil. For the three official profiles,

$$\log_2 \epsilon_{\text{seed}} \leq -1048.291, \qquad -1551.477, \qquad -2096.594. \tag{18}$$

The bound holds for every fixed hidden oil space; it does not average over key generation.

Proof. There are $(q^m - 1)/(q - 1)$ projective oil points. For any fixed projective point $[u]$, a uniform seven-dimensional subspace contains $[u]$ with probability $(q^7 - 1)/(q^n - 1)$. A union bound proves (17). The numerical values substitute the three official pairs (n, m). $\square$

Constructing the isotropic subspace. For a quadratic form Q over a field of odd characteristic, write

$$B_Q(u, z) = \frac{Q(u + z) - Q(u) - Q(z)}{2}$$

for its polar bilinear form. A subspace is *totally isotropic* for Q when Q vanishes identically on it.

Theorem 4.3 (Building a common isotropic subspace)**.** Let $Q_1, \dots, Q_s$ be homogeneous quadrics on $\mathbb{F}_q^n$, and let $0 \neq w$ satisfy $Q_i(w) = 0$ for every i. If

$$n \geq (s + 1)r + s, \tag{19}$$

then a deterministic finite algorithm constructs an r-dimensional subspace U containing w on which every Q_i vanishes identically.

At each extension step the algorithm uses linear algebra and examines at most

$$\left| \mathbb{P}^{2s}(\mathbb{F}_q) \right| = \frac{q^{2s+1} - 1}{q - 1} \tag{20}$$

projective points.

Proof. At each step, use linear equations to restrict to vectors z satisfying $B_{Q_i}(u, z) = 0$ for every current basis vector u and every selected form Q_i. Condition (19) leaves at least $2s + 1$ quotient dimensions, where Chevalley–Warning supplies a nonzero common zero of the induced s quadrics [30, 31]. Lifting this zero extends the isotropic subspace by one dimension. The complete dimension count appears in Section E. □

Corollary 4.4 (Affine reduction after target normalization). Let $P : \mathbb{F}_q^n \to \mathbb{F}_q^m$ be any homogeneous quadratic map, let $y \neq 0$, and choose $1 \leq s < m$. After an invertible output recombination sends y to the last coordinate, a nonzero common zero of any s of the first $m - 1$ transformed quadrics can be extended to an $(m - s)$-dimensional common totally isotropic subspace whenever

$$n \geq (s + 1)m - s^2. \tag{21}$$

On that subspace, the affine problem has $m - s$ equations in $m - s$ variables.

Proof. Apply Theorem 4.3 with $r = m - s$. Its condition becomes $n \geq (s + 1)(m - s) + s = (s + 1)m - s^2$. The selected s forms vanish identically on the constructed subspace and have zero target, leaving the stated affine residual. □

The affine formulation still counts a common scalar that does not change any homogeneous zero-target equation. A projective point $[z]$ records a nonzero vector z only up to multiplication by a nonzero scalar. The next corollary uses this viewpoint to fix the scale and remove one variable exactly.

Corollary 4.5 (Projective reduction after target normalization). Assume that q is odd. Under the hypotheses of Corollary 4.4, put $r = m - s$ and let U be the constructed r-dimensional subspace. Normalize the target to $(0, \dots, 0, 1)$ and write

$$H_1, \dots, H_{r-1}, H_0$$

for the restrictions to U of the remaining transformed forms, with H_0 corresponding to the nonzero target coordinate. Then:

1. affine solutions of
$$H_1(z) = \cdots = H_{r-1}(z) = 0, \qquad H_0(z) = 1 \tag{22}$$
modulo the pair $z \sim -z$ are in bijection with projective points $[z] \in \mathbb{P}(U)$ satisfying
$$H_1(z) = \cdots = H_{r-1}(z) = 0 \tag{23}$$
and $H_0(z)$ a nonzero square in $\mathbb{F}_q$;
2. a solution of (22) has invertible affine Jacobian if and only if the Jacobian of (23) has full rank at its projective point; and
3. after choosing a uniform nonzero linear form $L \in U^*$, the chart $L(z) = 1$ turns (23) into $r - 1$ quadrics in $r - 1$ affine variables. Any fixed projective root lies in this chart with probability
$$1 - \frac{q^{r-1} - 1}{q^r - 1} > 1 - \frac{1}{q}.$$

Thus the system has $m - s - 1$ equations in $m - s - 1$ variables. Bézout's theorem bounds its number of isolated roots by 2^{m-s-1}.

Proof. Homogeneity multiplies every quadratic value by λ^2 under $z \mapsto \lambda z$. Thus H_0 can be scaled to one exactly when its value is a nonzero square. Euler's identity shows that the affine Jacobian is invertible exactly when the projective Jacobian has full rank after removing the common scaling direction. Counting the linear forms that vanish at a fixed projective point gives the chart probability. The full Jacobian argument appears in Section E. □

For the QR-UOV parameters, set $s = 3$ and $r = m - 3$. The subspace condition is

$$n \geq 4m - 9. \tag{24}$$

The three official profiles give

$$(n, 4m - 9) = (210, 207), \quad (306, 303), \quad (411, 411).$$

The resulting projective solver dimensions are

$$k = r - 1 = m - 4 \in \{50, 74, 101\}. \tag{25}$$

The seed root and every extension step use finite exhaustive search, so this reduction needs no existence oracle. Even after overcharging the required linear algebra and search, preprocessing costs at most $2^{64.501}$ Boolean gates across the three profiles. The complete cost calculation appears in Section E.

4.2 Why the reduced system has a usable root

The structured degree-three conversion does not produce uniformly random quadratic forms over $\mathbb{F}_q$. We therefore prove the exact randomness property needed for the root count instead of treating the reduced equations as random. Let $E = K^{V+O}$, viewed over $\mathbb{F}_q$, and let $\widehat{\mathcal{O}} \subset E$ be the hidden base-field oil space. In central coordinates, each form in the random expanded-key model has the form

$$Q(v, o) = \mathrm{Tr}_{K/\mathbb{F}_q}\Big(\gamma(v^\top A v + 2v^\top B o)\Big) \tag{26}$$

for fixed nonzero $\gamma \in K$ and uniform $A \in \mathrm{Sym}_V(K)$ and $B \in K^{V \times O}$.

Lemma 4.6 (Randomness of values and first derivatives)**.** Let $x \in E \setminus \widehat{\mathcal{O}}$, and let $U \subseteq E$ contain x. For a uniform form Q from (26), $Q(x)$ is uniform in $\mathbb{F}_q$. Conditioned on $Q(x) = a$, the derivative $dQ_x|_U$ is uniform among the linear forms $\lambda \in U^*$ satisfying $\lambda(x) = 2a$. Moreover, evaluation at two off-oil points is linearly dependent only when the points differ by a nonzero scalar in $\mathbb{F}_q$.

Proof. Nondegeneracy of the trace pairing lets the central bilinear form prescribe every value and derivative allowed by Euler's identity. Comparing the rank-one coefficient matrices at two points gives the last claim. The full calculation of all possible values and derivatives, including the conversion from the degree-three field, appears in Section E. □

Let U be the subspace from Theorem 4.3, and choose an $\mathbb{F}_q$-basis to identify U with $\mathbb{F}_q^r$. Under the normalization (16), write

$$H_j = Q_{3+j}|_U \quad (1 \leq j \leq r - 1), \qquad H_0 = Q_m|_U.$$

The remaining affine system is

$$H_1(z) = \cdots = H_{r-1}(z) = 0, \qquad H_0(z) = 1, \qquad r = m - 3. \tag{27}$$

The projective solver uses only the first $r - 1$ equations; H_0 filters and scales their projective roots.

The next statement gives an explicit success probability rather than assuming that a system which looks random has a root. Let

$$\pi_{r-1}(q) = \prod_{j=1}^{r-1} (1 - q^{-j}). \tag{28}$$

Proposition 4.7 (Probability of an affine root with invertible Jacobian)**.** Condition on the selected forms, the public randomness used to construct U, and an off-oil seed point. Put

$$d = \dim_{\mathbb{F}_q}(U \cap \widehat{\mathcal{O}}), \qquad \mu = 1 - q^{d-r}.$$

Then $d \leq r - 1$. Over the independent remaining forms in the random expanded-key model, the probability that (27) has an $\mathbb{F}_q$-rational root with invertible $r \times r$ Jacobian is at least

$$\frac{\mu\, \pi_{r-1}(q)^2}{2 + \mu - (q-1)q^{-r}}. \tag{29}$$

For $q = 127$ and each of $r = 51, 75, 102$, this is at least

$$p_* = 0.326336998767509\ldots. \tag{30}$$

Proof. The off-oil seed point gives $d \leq r - 1$. Let M count all off-oil roots of (27). Lemma 4.6 gives $\mathbb{E}[M] = \mu$. Pairing proportional and nonproportional points gives

$$\mathbb{E}[M^2] = 2\mu + \mu^2 - (q-1)\mu q^{-r}. \tag{31}$$

At a fixed root, the same lemma makes the Jacobian invertible with probability $\pi_{r-1}(q)$. If N counts these roots, then $\mathbb{E}[N] = \mu\pi_{r-1}(q)$ and $N^2 \leq M^2$. The second-moment inequality now gives (29); using $\mu \geq 1 - q^{-1}$ gives (30). The complete pair count appears in Section E. $\square$

Corollary 4.8 (Probability of a usable projective root)**.** Under the same conditioning, the probability that the projective system

$$H_1 = \cdots = H_{r-1} = 0 \quad \text{in } \mathbb{P}^{r-1}(\mathbb{F}_q)$$

has a point $[z]$ with full-rank projective Jacobian for which $H_0(z)$ is a nonzero square is at least the bound in (29), and hence at least p_* for every official profile.

Proof. By Corollary 4.5, every affine root of (27) with invertible Jacobian maps to a projective root with full-rank Jacobian and the required nonzero-square value; the two affine roots z and $-z$ map to the same projective point. Existence of an affine root therefore implies the displayed projective event. $\square$

The factor $1/p_* = 3.0643169600\ldots$ adds 1.615566 bits to the expected cost. A random chart adds less than $\log_2(127/126) = 0.011405$ bits. The root bound is proved from the QR-UOV distribution by a second-moment argument. The probability that the initial subspace avoids the oil space holds for every fixed key. The later root probability averages over the independent-graph model. We do not claim that repeated attacks on every fixed seeded key behave independently.

4.3 The verified attack

A projective chart fixes the common scale by choosing a nonzero linear form L and requiring $L = 1$. With this terminology, the complete direct attack is as follows. The root solver also applies a random linear form to label its candidate roots. We call this form a *separator* when the roots that the algorithm must distinguish receive different labels.

Theorem 4.9 (Verified QR-UOV direct forgery)**.** Work in the independent-graph random expanded-key model and the official degree-three pull-back. For any nonzero target y, the following public algorithm has one-sided error: it may fail to find a forgery, but it never returns an invalid one.

1. Normalize the target to $(0, \ldots, 0, 1)$ as in (16).
2. Choose a uniform seven-dimensional seed subspace, enumerate its projective points, and take a nonzero common zero of the first three forms.
3. Use Theorem 4.3 to construct an $(m-3)$-dimensional common totally isotropic subspace for those forms.
4. Choose a uniform nonzero chart form L on that subspace. Substitute $L = 1$ in the $m-4$ remaining zero-target quadrics, and apply Theorem 7.1 to the resulting square system in

   $$k = m - 4$$

   variables.
5. For each $\mathbb{F}_q$-rational root returned by the solver, recover its projective point, test whether the last quadratic value is a nonzero square, scale to make that value one, reverse the public transformations, and evaluate the unchanged public map.

Every returned vector is a valid signature preimage. Over the random expanded-key model and the attack's public randomness, one complete invocation reaches a reduced system with a root over $\mathbb{F}_q$ whose Jacobian is invertible and whose last quadratic value is a nonzero square with probability at least

$$(1 - \epsilon_{\text{seed}})p_* \left(1 - \frac{1}{q}\right)\left(1 - \frac{2^{2k}}{|k'|}\right), \qquad k = m - 4, \tag{32}$$

where $k'/\mathbb{F}_q$ is the solver working extension. Taking

$$[k' : \mathbb{F}_q] = 15, \qquad 22, \qquad 30$$

at Levels I, III, and V gives the finite costs reported in Section 10. We multiply the cost of one invocation by the inverse of (32). This is a conservative expected-attempt factor, not a claim that retries on every fixed key are independent.

Proof. The first $m-1$ target coordinates are zero. By Proposition 4.2, the seed subspace is disjoint from the hidden oil space with probability at least $1 - \epsilon_{\text{seed}}$ for every fixed key. Chevalley–Warning then supplies an off-oil seed root, and Theorem 4.3 constructs the advertised subspace. Conditional on the selected first three forms, the graph, the public randomness, and this off-oil seed point, the remaining transformed forms are still independent uniform pull-back forms. Corollary 4.8 supplies a usable projective root with full-rank Jacobian with probability at least p_*. A random nonzero chart contains a fixed such root with probability greater than $1 - 1/q$. The solver of Section 7 includes every root in the chart with the separator probability in (32). Square testing and scaling recover the associated affine solutions exactly. Re-evaluation of all m original public forms rejects every invalid candidate, so the algorithm cannot output an invalid signature. $\square$

4.4 Why changing only the vinegar count cannot help

Combining the direct reduction with equivalent-key recovery gives a bound for every vinegar dimension. Let the quotient degree be ℓ, so that

$$n = \ell V + m.$$

Applying Corollary 4.4 with a general value of s gives the required $(m-s)$-dimensional common-isotropic subspace whenever

$$\ell V \geq s(m-s). \tag{33}$$

By Corollary 4.5, the corresponding nonzero-target square system has $m-s-1$ variables.

Theorem 4.10 (Fixed-output bound across all vinegar dimensions)**.** Work over a field of odd characteristic. Fix m and quotient degree ℓ. For every nonzero target, within the two attack families analyzed in this paper a square system in at most $m-\ell-1$ variables is available for every integer vinegar count V:

$$r_{\text{attack}}(V) \leq m-\ell-1. \tag{34}$$

More precisely:

1. if $V \leq m-\ell-1$, equivalent-key recovery uses a square system in V variables;
2. if $V \geq m-\ell$, the direct reduction with $s=\ell$, followed by fixing the common scale, uses a square system in $m-\ell-1$ variables.

These cases cover every integer V. For QR-UOV, $\ell=3$, so the ceiling is $m-4$.

Proof. In the first case, $V<m$, so the m key-recovery equations admit a V-equation square recombination and the planted-root attack of Section 5 has at most $V \leq m-\ell-1$ variables. In the second case,

$$\ell V \geq \ell(m-\ell),$$

which is (33) with $s=\ell$. The affine direct system has dimension $m-\ell$ and Corollary 4.5 removes the common-scale coordinate, leaving $m-\ell-1$ variables. Since consecutive integers $m-\ell-1$ and $m-\ell$ meet the two inequalities, no case is omitted. □

The theorem bounds the number of variables in the nonlinear system. It does not claim that running time depends only on this number. For the official degree-three sets, Proposition 4.2 and Corollary 4.8 supply the success factors charged in the concrete cost calculation. Its design implication is exact: once V reaches $m-\ell$, increasing V while holding m fixed no longer enlarges the projective direct-forgery system. Conversely, decreasing V into the other case makes key recovery no larger than the same ceiling. Moving both attack lines therefore requires changing m as well as V, or modifying the construction so that one reduction no longer applies.

The direct construction and the complete attack have both been executed on reduced parameters. The exact traces and public replay checks appear in Sections E and K; these are correctness tests, not full-parameter benchmarks.

5 Equivalent-key recovery from a planted root

The second attack uses a root guaranteed by the hidden key structure. Such a guaranteed but unknown root is often called a *planted root*. Fixing an oil vector $c \in K^O$ turns the hidden graph point $(-Gc, c) \in \mathcal{O}$ into the planted root Gc of public quadratic equations in V variables. If the matrix of first derivatives has full rank at that root, public linear algebra recovers every column of G. The attack then checks (3), applies the pull-back (7), verifies the factorization (8), and finds a signing witness for (13). The end of this section records a second way to recover the same oil space from the derivatives at a projective oil point.

5.1 Public equations with a hidden root

Fix $c \in K^O$. Substitute the public oil coordinate c and the negated vinegar coordinate $-x$ into each public form:

$$q_{i,c}(x) := P_i(-x, c) = x^\top A_i x - 2x^\top E_i c + c^\top C_i c, \qquad x \in K^V. \tag{35}$$

The equations $q_{1,c} = \cdots = q_{m,c} = 0$ are public. Because the choice of c fixes one direction in the oil coordinates, we call these the *fixed-direction equations*. Define their public derivative matrix by

$$\mathcal{D}_c(x) := \begin{pmatrix} (E_1 c - A_1 x)^\top \\ \vdots \\ (E_m c - A_m x)^\top \end{pmatrix}. \tag{36}$$

Recall that $B_i = E_i - A_i G$, and let $\mathcal{B}(c)$ have ith row $(B_i c)^\top$.

Proposition 5.1 (A planted root for every direction)**.** For every $c \in K^O$, the point $x_c = Gc$ is a common zero of (35). More precisely, for every $z \in K^V$,

$$q_{i,c}(Gc + z) = z^\top A_i z - 2(B_i c)^\top z. \tag{37}$$

Consequently,

$$\mathcal{D}_c(x_c) = \mathcal{B}(c), \qquad J_q(x_c) = -2\mathcal{B}(c).$$

Proof. Expand (35) at $x = Gc + z$. The graph identity (3) makes the constant term zero, while the linear term is $-2(B_i c)^\top z$. This proves (37); differentiation gives the two matrix identities. □

5.2 A root with full-rank derivative determines the graph

A root x for one oil vector c gives only one graph value. When the derivative matrix has full column rank, the same public equations determine the graph value on every other oil vector and therefore determine all of G.

Proposition 5.2 (Exact graph completion)**.** Let x be a common zero of all equations $q_{i,c}$ and suppose $\operatorname{rank} \mathcal{D}_c(x) = V$. For $d \in K^O$, define

$$b_{c,d}(x)_i := c^\top C_i d - x^\top E_i d \qquad (1 \le i \le m). \tag{38}$$

Then the public system

$$\mathcal{D}_c(x) y = b_{c,d}(x) \tag{39}$$

has at most one solution $y \in K^V$. If $x = Gc$, its solution is $y = Gd$. Solving (39) for the O coordinate vectors $d = e_j$ therefore recovers G exactly. Checking every identity (3) then certifies a valid expanded UOV decomposition.

Proof. Full column rank gives uniqueness. For $x = Gc$, multiply (3) by $c^\top$ on the left and d on the right. The result is

$$(E_i c - A_i G c)^\top G d = c^\top C_i d - (Gc)^\top E_i d,$$

which is row i of (39). Hence Gd is the unique solution. The verified-key conclusion follows from Proposition 3.4. □

Remark 5.3 (A compact attack certificate). A successful candidate is certified by (c, x, G) and one full-rank minor of $\mathcal{D}_c(x)$. A verifier checks the fixed-direction equations, the graph completion systems, all graph identities, the inverse pull-back, and a signing witness. The certificate makes no claim about other roots or solver completeness.

5.3 How often the derivative has full rank

We now bound the probability that the planted root passes the full-rank test. The following quantity is the exact full-rank probability for a uniform $m \times V$ matrix over K. Let

$$\rho_{m,V}(Q) := \prod_{a=0}^{V-1} (1 - Q^{a-m}), \qquad \tau_{m,V}(Q) := 1 - \rho_{m,V}(Q). \tag{40}$$

Thus $\rho_{m,V}(Q)$ is the probability that a uniform $m \times V$ matrix over K has full column rank.

Proposition 5.4 (Rank of the planted derivative)**.** For a fixed expanded key, suppose some maximal minor of $\mathcal{B}(c)$ is a nonzero polynomial in c. A fresh uniform $c \leftarrow K^O$ then satisfies

$$\Pr[\operatorname{rank} \mathcal{B}(c) = V] \geq 1 - \frac{V}{Q}. \tag{41}$$

In the random expanded-key model, $\mathcal{B}(c)$ is a uniform $m \times V$ matrix for every fixed nonzero c. Moreover, the coordinate-direction matrices $\mathcal{B}(e_1), \dots, \mathcal{B}(e_O)$ are independent, so

$$\Pr\big[\operatorname{rank} \mathcal{B}(e_j) < V \text{ for every } j\big] = \tau_{m,V}(Q)^O. \tag{42}$$

Proof. Every maximal minor of $\mathcal{B}(c)$ is homogeneous of degree V, so Schwartz–Zippel gives (41). Under Definition 3.1, each B_i is an independent uniform linear map. This gives the fixed-direction law and independence across coordinate directions. The exact rank calculations appear in Section D. $\square$

The verified attack later uses a fresh random linear recombination to select V of the m equations. The supplement also gives a deterministic public list of only $V(m-V)+1$ recombinations, together with exact list sizes and failure probabilities for the official parameters (Section D).

5.4 An alternative recovery from a projective oil point

The supplement gives a separate route that finds a nonzero point x on the oil space, considered only up to a common scalar (Section A). Let L/K be the field over which this point is found. From x, form the public matrix

$$C_x = (P_1 x \ \cdots \ P_m x) \in L^{(V+O)\times m}.$$

Lemma 5.5 (The common derivative kernel)**.** If $x \in \mathbb{P}(\mathcal{O}_L)$, then $\mathcal{O}_L \subseteq \ker C_x^{\mathsf{T}}$. If $\operatorname{rank} C_x = V$, equality holds.

Proof. For $u \in \mathcal{O}_L$, (6) gives $u^{\mathsf{T}} P_i x = 0$ for every i. If $\operatorname{rank} C_x = V$, the kernel of its transpose has dimension $(V+O) - V = O$. $\square$

When the derivatives at x have rank V, the lemma recovers the entire oil space as a kernel. Row reduction, recovery of a basis over K, and the checks from Propositions 3.4 and 3.7 then produce a verified signing key. The attack does not need the oil space to be unique: it keeps every exactly verified candidate with a signing witness. Section H separately bounds the probability of a second K-rational oil space.

6 Why one root with invertible Jacobian is enough

The exact solver used below comes with algebraic conditions on the complete solution set. At first sight, these conditions are stronger than what the attack can prove. The attack knows only that the planted root has an invertible Jacobian. This section shows that the solver may restrict its attention to the set where the Jacobian is invertible, so singular roots and components elsewhere are irrelevant.

Fix $c \in K^O$ and combine the m public equations into V equations using a public matrix $R \in K^{V \times m}$. Denote the resulting equations by

$$f = (f_1, \ldots, f_V)^\mathsf{T} := R(q_{1,c}, \ldots, q_{m,c})^\mathsf{T}$$

and let h be the determinant of their Jacobian:

$$h(x) := \det J_f(x). \tag{43}$$

Working only where $h \neq 0$ is called *localization at h*: algebraically, we allow division by powers of h. This removes every point where the Jacobian is singular. On the remaining set, the equations satisfy the two technical conditions required by the solver.

Lemma 6.1 (Solver conditions where the Jacobian is invertible)**.** Let k be a perfect field, let $f_1, \ldots, f_V \in k[x_1, \ldots, x_V]$, and let $h = \det J_f$. In the localization

$$S_h := k[x_1, \ldots, x_V]_h,$$

the sequence $f_1, \ldots, f_V$ is regular. For every $1 \leq s \leq V$, the ideal $(f_1, \ldots, f_s)S_h$ is radical.

Here regular means that the equations cut the dimension by one at each step. Radical means that their common zeros carry no repeated algebraic multiplicity. The lemma applies these statements only on the set $h \neq 0$.

The proof appears at the start of the finite-field solver appendix in the technical supplement. It applies the Jacobian criterion to each prefix and then uses the complete-intersection criterion in S_h.

If $\mathcal{B}(c)$ has full column rank and $R\mathcal{B}(c)$ is invertible, Proposition 5.1 gives

$$h(Gc) = (-2)^V \det(R\mathcal{B}(c)) \neq 0. \tag{44}$$

Corollary 6.2 (The solver conditions hold near the planted root)**.** If $\mathcal{B}(c)$ has full column rank and $R\mathcal{B}(c)$ is invertible, then Gc belongs to the set defined by the equations and inequality

$$\{f_1 = \cdots = f_V = 0,\ h \neq 0\}.$$

The input tuple $(f_1, \ldots, f_V, h)$ satisfies hypotheses (H1)–(H2) of the solver of Giménez et al. [15], which accepts equations together with the inequality $h \neq 0$. These conclusions are independent of every root and algebraic component contained in $h = 0$.

Proof. Equation (44) puts the planted root in the open set. Lemma 6.1 gives exactly the two algebraic conditions required after localization. □

Thus the attack needs an invertible Jacobian at the planted root. It does not need the Jacobian to be invertible at every root or on every component of the full system.

7 Finding roots over the QR-UOV base field

The equivalent-key attack needs one isolated root at which the Jacobian is invertible. It does not need all roots or a description of every algebraic component. We use the exact symbolic solver of Safey El Din and Schost for this task [24]. The solver connects an easy system whose roots are known to the target equations and follows each root as the connecting parameter changes. We call the path followed by one root a *branch*. This section adapts the solver to characteristic 127 and states exactly what it returns.

7.1 The roots sought by the solver

All finite fields are perfect, which allows the Jacobian to detect repeated roots. Let k be any perfect field and let

$$f = (f_1, \dots, f_V) \in k[X_1, \dots, X_V]^V$$

be a square system of affine quadrics. Write

$$Z_{\mathrm{ns}}(f) := \{x \in \overline{k}^V : f(x) = 0,\ \det J_f(x) \neq 0\}. \tag{45}$$

Every point in $Z_{\mathrm{ns}}(f)$ is isolated. Bézout's theorem gives

$$C := 2^V, \qquad C_0 := 2^V + V2^{V-1} = 2^{V-1}(V+2). \tag{46}$$

Here C bounds the number of isolated roots. The solver represents the paths from the easy system to the target as an algebraic curve in one connecting parameter, and C_0 bounds the degree of that curve. Recovering a rational function of degree C_0 requires power-series coefficients through precision $2C_0$. This precision determines the main cost below. The supplement gives the multihomogeneous degree calculation that yields the formula for C_0.

The published algorithm starts from a product of univariate equations with known simple roots and follows at most C branches to the target equations. It then discards outputs whose target Jacobian is not invertible. The algorithm also uses the separator introduced in Section 4 to assign a scalar value to each root. The published proof uses a lower bound on the field characteristic only to construct the starting equations and this separator. We replace both constructions.

7.2 Why the solver works over the QR-UOV base field

Safey El Din and Schost's published characteristic bound enters through two finite families: the easy starting system and the separator [24]. We replace the start by a Vandermonde product system with exactly 2^V explicit simple roots, which requires only $2V$ distinct field elements. We replace the integer-indexed separator family by a uniform linear form. At most C^2 choices cause two roots to receive the same value, so a uniform choice works with probability at least $1 - C^2/|k'|$.

All remaining steps use exact field operations and power-series lifting. They do not divide by factorials or by any value that could vanish specifically in characteristic 127. The technical supplement gives both constructions, the lifting identities, and a step-by-step comparison with the published algorithm.

Theorem 7.1 (Roots with invertible Jacobian in characteristic 127)**.** Let k be a finite field and let $f = (f_1, \dots, f_V)$ be affine quadrics over k, represented by an arithmetic circuit of size L. Let k'/k be a finite extension containing at least $2V$ elements. There is a randomized algorithm with the following properties. Here $\widetilde{O}$ suppresses factors that are polynomial in logarithms of the displayed quantities.

1. Every returned point belongs to $Z_{\mathrm{ns}}(f)$.

2. Every point of $Z_{\mathrm{ns}}(f)$ is included in the output representation with probability at least

$$1 - \frac{C^2}{|k'|}. \tag{47}$$

3. The arithmetic cost is

$$\widetilde{O}\Big(CC_0(L + 2V + V^2)V\Big) \tag{48}$$

operations in k'.

In particular, $|k'| \geq 8C^2$ gives an inclusion probability of at least $7/8$ in one invocation.

Proof. Run the symbolic solver with the Vandermonde start and uniform separator described above. The construction and proof crosswalk in the technical supplement show that the remaining steps introduce no further characteristic bound. The separator succeeds with probability at least (47); on that event the published specialization recovers every isolated target root with invertible Jacobian. Its final Clean step removes every point with singular target Jacobian, even after a bad separator.

For V quadrics in one block, the degree quantities in the published Proposition 5 are exactly C and C_0 from (46), and the sum of degrees is $2V$. The new starting system costs $O(V^2C)$ and does not change the dominant lifting term. Applying that proposition gives (48). □

Remark 7.2 (The attack needs validity, not a complete root list)*.* The original solver paper notes that it cannot cheaply certify that a generic output contains every root. QR-UOV does not need such a certificate. We check every original fixed-direction equation, require the root coordinates to lie in K, complete the graph, and verify the unchanged public map. An omitted planted root causes a restart. A candidate that is not a valid root or key is rejected.

7.3 Complete equivalent-key attack

We now combine the fixed-direction equations, the root solver, graph completion, and the public signing test into one attack.

For a random $V \times m$ matrix R over K, put

$$f_{c,R} := R(q_{1,c}, \ldots, q_{m,c})^{\mathsf{T}}.$$

If $\mathcal{B}(c)$ has rank V, then $R\mathcal{B}(c)$ is a uniform $V \times V$ matrix. Write

$$\mu_V(Q) := \prod_{j=1}^{V}(1 - Q^{-j}), \qquad Q = |K| = 127^3. \tag{49}$$

Choose $K' = \mathbb{F}_{Q^r}$ with

$$r := \min\{s \geq 1 : Q^s \geq 8 \cdot 2^{2V}\}. \tag{50}$$

The values are

$$r = 6, \qquad 8, \qquad 10 \tag{51}$$

at Levels I, III, and V.

Theorem 7.3 (Verified equivalent-key recovery)**.** Let a fixed expanded QR-UOV public key have a valid graph decomposition G. Suppose

1. some maximal minor of $\mathcal{B}(c)$ is a nonzero polynomial in c; and
2. the recovered planted decomposition has a nonzero signing determinant polynomial Δ from Proposition 3.7.

Then the following algorithm eventually returns another signing key for the same public key. It never returns an incorrect key and repeats with fresh randomness when an attempt fails.

1. Sample $c \leftarrow K^O$ and $R \leftarrow K^{V \times m}$.
2. Run Theorem 7.1 on $f_{c,R}$ over K' from (50).
3. Keep only roots that solve all m original fixed-direction equations and whose coordinates lie in K.
4. For every retained root x, run the graph completion of Proposition 5.2. Require all graph identities, the inverse pull-back, exact public-map factorization, and a signing witness.
5. Return the first accepted key; otherwise restart with fresh public coins.

One invocation includes the planted branch with probability at least

$$p_{\mathrm{dir}} \geq \left(1 - \frac{V}{Q}\right) \mu_V(Q) \left(1 - \frac{2^{2V}}{Q^r}\right). \tag{52}$$

The expected numbers of full solver invocations are at most

$$1.0000262, \qquad 1.0000561, \qquad 1.0202201. \tag{53}$$

Every returned key is correct, regardless of whether failed solver invocations included every root.

Proof. By Proposition 5.4, a random direction gives a full-rank derivative at the planted root with probability at least $1 - V/Q$. Conditional on that event, $R\mathcal{B}(c)$ is a uniform square matrix, so it is invertible with probability $\mu_V(Q)$. Proposition 5.1 then makes the planted point an isolated root of $f_{c,R}$ with invertible Jacobian. Theorem 7.1 includes it with probability at least $1 - 2^{2V}/Q^r$.

The filters for the original equations and for coordinates in K retain the planted point. Proposition 5.2 reconstructs G, and Propositions 3.4 and 3.7 supply the exact factorization and signing certificate. Conversely, the algorithm rejects a candidate that fails any original equation, graph completion equation, graph identity, pull-back identity, factorization check, or oil-matrix test. Thus every accepted key is correct. Retrying with fresh randomness and using (52) gives the numerical expectations above. □

Corollary 7.4 (Probability that all public choices fail)**.** In the random expanded-key model, the probability that every coordinate direction has deficient planted derivative is $\tau_{m,V}(Q)^O$, with base-two logarithms

$$-1132.166907, \qquad -1635.352198, \qquad -2935.247544.$$

The probability that the deterministic signing search over pairs of basis matrices fails has base-two logarithm below

$$-544.820, \qquad -796.276, \qquad -1068.687.$$

Outside their union, a deterministic scan of coordinate directions and signing witnesses includes a successful choice. Only the internal root solver requires randomness.

Proof. Combine Propositions 5.4 and 3.7. The two exceptional events need not be independent, so a union bound suffices. □

8 Concrete cost of the symbolic solver

The preceding theorem gives an asymptotic cost, but a concrete estimate also depends on how the solver represents and multiplies large polynomials. This section states the resulting estimate first. It then explains the choices that change the final number: evaluating all quadratic equations together, using extension fields with sparse defining polynomials, distinguishing only the root that may yield a forgery, and testing exactly whether that root lies in the base field. The section ends with the success probability and the complete Level-I cost.

8.1 Resulting conditional cost estimate

Let $q = 127$, $a = m - 4$, $C = 2^a$, and $C_0 = 2^{a-1}(a + 2)$ as in the direct attack. We compute the cost for several extension fields whose sparse defining polynomials are proved irreducible, then select the least expensive degree.

Theorem 8.1 (Conditional Boolean-gate estimate)**.** Assume the random expanded-key model, the degree bound for the curve followed by the solver, the polynomial multiplication costs stated below, and the specification's Boolean-gate convention. Require the separator to distinguish only a root that passes the forgery tests, as justified in Section 8.5, and use the exact base-field test in Section 8.6. After accounting for failed attempts, the estimated costs are those in Table 1. In particular, Level I costs

$$2^{140.383995063}$$

Boolean gates, leaving 2.616004937 bits below the category target.

Table 1: Estimated base-two logarithms of Boolean-gate costs. "All roots" requires the separator to distinguish every root. "Accepted root" requires it only for a root that passes the forgery tests.

Level	Original	All roots	Accepted root	Target	Margin
I	151.639	141.126	140.384	143	+2.616
III	203.507	191.630	191.027	207	+15.973
V	260.389	247.637	247.125	272	+24.875

The least expensive Level-I row uses $K' = \mathbb{F}_{127^8}$. The Rabin irreducibility test certifies the following defining polynomial:

$$X^8 + X^4 - 1.$$

The nearby degree-9 choice $X^9 + X^2 + 1$ gives 140.418965, which provides a cross-check on the optimization.

8.2 Batching equation evaluation and polynomial multiplication

The V dense quadrics share one vector of quadratic monomials. Evaluating that vector once and using it for all V equations removes a redundant factor from the arithmetic-circuit cost. To follow the roots, the solver stores truncated power series in a parameter T. A second variable U labels the roots through the separator. At precision k, these coefficients lie in

$$\mathcal{A}_k = K'[T, U]/(T^k, Q_k(U)), \qquad \deg_U Q_k \le C.$$

Here Q_k is the monic polynomial whose roots encode the current root values. We encode the pair of exponents (i, j) as one exponent, using radix $2C - 1$. This converts multiplication in $\mathcal{A}_k$ into three ordinary polynomial products, including reduction modulo Q_k. Let $\mathsf{M}(t)$ denote the cost of multiplying polynomials of length t. Across all stages that double the precision, one product in $\mathcal{A}_k$ is bounded by

$$3\mathsf{M}(8CC_0). \tag{54}$$

The factor 8 accounts for the final precision $2C_0$. The technical supplement gives the batched circuit, the exponent encoding, reduction modulo Q_k, and the proof of the resulting asymptotic cost $\widetilde{O}(V^{\omega+1}4^V)$.

8.3 Sparse working fields

The solver performs arithmetic in an extension of $\mathbb{F}_{127}$. A sparse defining polynomial reduces the number of base-field operations needed after each multiplication. The direct solver may use any extension degree. The equivalent-key solver starts over $K = \mathbb{F}_{127^3}$, so its degree over $\mathbb{F}_{127}$ must be divisible by three. Rabin tests certify sparse irreducible polynomials of degrees $15, 22, 30$ for direct forgery and $18, 24, 30$ for equivalent-key recovery. The refined direct estimate separately selects degrees $8, 12, 16$.

For an extension degree d, let (M_d, A_d) be the audited counts of base-field multiplications and additions in the selected recursive Karatsuba circuit, and let s_d count the nonleading terms of the sparse modulus. One multiplication in the working field is charged by the following upper bound:

$$\mathsf{G}^{\text{sparse}}_{127^d} = M_d \mathsf{G}_{\text{mul}}(\log_2 127) + \big(A_d + s_d(d-1)\big) 4 \log_2 127. \tag{55}$$

Here $\mathsf{G}_{\text{mul}}(b) = 2b^2 + b$ is the specification's gate charge for multiplying two b-bit values. The supplement gives every defining polynomial, its irreducibility certificate, the recursive operation count, and a direct multiplication cross-check.

8.4 Collisions among rejected roots do not raise the degree

This subsection explains why only the root that eventually passes the forgery tests needs a unique scalar label. Collisions among roots that will be rejected need not increase the solver's cost.

As the connecting parameter t changes, the solver represents all moving roots at once in a finite-dimensional algebra. After removing components that never reach the target equations, the corresponding curve has degree at most $D = C_0$. Introduce formal coefficients Λ_j and assign each root the value $\sum_j \Lambda_j x_j$. The characteristic polynomial $\chi_\Lambda(t, U)$ has these values as its roots. A concrete linear form λ may assign the same value to two roots, but the following lemma shows that such a collision does not increase the degree in t.

Lemma 8.2 (A collision does not increase the degree)**.** For every $\lambda \in K'^a$, including a nonseparating choice,

$$q_\lambda(t, U) = \chi_\Lambda(t, U)|_{\Lambda=\lambda}$$

has a common-denominator representation whose numerator and denominator have t-degree at most D.

Lemma 8.3 (Recovering coordinates by differentiation)**.** The interpolation numerator for coordinate x_j is

$$v_j(t, U) = -\left.\frac{\partial \chi_\Lambda(t, U)}{\partial \Lambda_j}\right|_{\Lambda=\lambda}.$$

It has the same degree bound. The numerator for any fixed linear combination of the coordinates also has degree at most D. The attack additionally uses the rational function $H_0/(\delta + \lambda)$ to test the final target equation. After adjoining this function as another coordinate, its numerator has degree at most $2D$.

Schost's parametric degree theorem gives these statements first for the generic linear form [25]. Specializing Λ can create repeated factors, but it cannot increase the degree in t because the common denominator is independent of Λ. The attack may therefore keep the full characteristic polynomial and require λ to distinguish only the root it accepts. The additional rational test gives reconstruction precision $4C_0 + 1$. The concrete cost still assumes that Schost's degree theorem applies after removing the irrelevant components of the exact attack curve and after adding this rational test.

8.5 Only the accepted root needs a unique label

When the connecting parameter reaches the target equations, the interpolated polynomials may share a power of that parameter. Cancel this common power. Let $\bar{q}$ be the resulting univariate polynomial whose roots label target candidates, and let $\bar{v}_b$ be the corresponding numerator for a scalar value b. The exponent of the lowest nonzero power of the connecting parameter is called its *valuation*. Some branches may diverge as the target is approached. The linear form λ must preserve the lowest nonzero power on those branches so that canceling the common power does not remove a finite target root.

Lemma 8.4 (Recovering one uniquely labeled root)**.** If λ preserves the minimum valuation of every branch escaping to infinity and τ is a simple root of $\bar{q}$, then exactly one finite represented point P satisfies $\lambda(P) = \tau$, and

$$b(P) = \frac{\bar{v}_b(\tau)}{\bar{q}'(\tau)}.$$

No separation condition is needed at any other root value.

For a fixed usable target point, at most $C - 1$ linear conditions on a uniform $\lambda \in (K')^a$ can violate the valuation condition or collide with that point. Hence

$$\Pr[\text{bad target separator}] \leq \frac{C-1}{|K'|}. \tag{56}$$

The rational test $H_0/(\delta + \lambda)$ uses a random scalar δ in its denominator. After rejecting shifts that vanish at a known starting root, its remaining denominator failure probability is

$$\Pr[\text{bad target denominator}] \leq \frac{C}{|K'| - C}. \tag{57}$$

The earlier analysis required one linear form to distinguish every pair among at most C roots, which cost $C^2/|K'|$. The supplement proves the weaker condition above and both failure bounds.

8.6 One power test identifies base-field roots

The solver works over $K' = \mathbb{F}_{q^d}$, but a QR-UOV forgery needs coordinates in the base field $\mathbb{F}_q$. For a candidate point P, set $b_1(P) = \lambda^{q^{d-1}} \cdot P$.

Lemma 8.5 (Exact base-field test for a simple root)**.** If $\tau = \lambda(P)$ is a simple root of $\bar{q}$, then

$$b_1(P)^q = \tau \quad \Longleftrightarrow \quad P \in \mathbb{F}_q^a.$$

Indeed, the equality says $\lambda(P^q) = \lambda(P)$. Raising every coordinate to its qth power maps target roots to target roots. If $P^q \neq P$, these two distinct roots would receive the same value under λ, making τ a repeated root. The test therefore accepts exactly the candidates in $\mathbb{F}_q^a$ and needs no second linear combination of the coordinates.

8.7 Root success probability

Let M count all off-oil affine roots and N those with invertible Jacobian. The exact first and second moments, together with the pairing $z, -z$, give

$$\Pr[\text{usable projective root with full-rank Jacobian}] \geq \frac{\mu\pi}{2} - \frac{\mu^2}{8} + \frac{(q-1)\mu q^{-r}}{8},$$

where $\mu \geq 1 - q^{-1}$ and $\pi = \prod_{j=1}^{r-1}(1 - q^{-j})$. Substitution yields

$$p_{\text{root}} = 0.3690869822\ldots.$$

Using instead the earlier proved bound $0.326336998767509\ldots$ raises the Level-I total only to 140.531461 bits, still 2.469 bits below target. The supplement retains the moment calculation.

8.8 Level-I cost calculation

At the selected extension degree $d = 8$, one attempt succeeds with probability at least 0.353997. The cost therefore includes an expected 2.824886 attempts. The failure probabilities for distinguishing the accepted root and for the denominator in its rational test are at most 0.016637 and 0.016918. We charge the full power-series lifting schedule, two sets of scalar interpolation data, one coordinate reconstruction, factorization, and public verification. We also add the cost of 64 long polynomial products for lower-order work. The total is

$$2^{140.383995063}$$

Boolean gates. The result stays below 2^{143} when the cost assigned to rational reconstruction ranges from 8 to 128 times its baseline. It also stays below target if the assumed cost of the long polynomial products is multiplied by eight, but not by sixteen. The technical supplement gives every component and both sensitivity tables.

9 Evidence and limitations

The two reductions and the fixed-output theorem are exact algebraic results. The same is true of the identities used to recover one distinguished root and to test whether its coordinates lie in the base field. None of these statements depends on the Boolean-gate cost model. The attack checks every candidate against the original equations or the unchanged public verifier. A failed solver call may force another attempt, but it cannot make the verifier accept an invalid signature or key.

We implemented the direct-forgery attack on smaller parameters. The program starts from expanded public matrices, a message, and a salt. It produces a forged signature and verifies it using only public data. Six instances over $\mathbb{F}_{13}$ and one over $\mathbb{F}_{29}$ completed successfully, and a separate public replay reproduced each result. The program keeps the QR-UOV quadratic map but simplifies the conversion from seeds, the byte encoding, and the hash interface. It also uses straightforward power-series arithmetic rather than the fast multiplication assumed by the full cost estimate. The experiment therefore checks the attack from the expanded key through verification. It does not test the official input encoding or the full-parameter running time (Section K).

The full-parameter cost estimate still depends on four unproved modeling steps:

1. Schost's degree theorem must apply to the exact attack curve after irrelevant components are removed and the rational target test is added.
2. The long polynomial products must admit a complete implementation with the cost assumed in Section 8.2.

3. The random expanded-key model must accurately describe the official seeded key expansion.
4. The implementation must move and store its polynomial coefficients, a cost omitted by the Boolean-gate count and already prohibitive at the stated memory sizes.

The Level-I margin is only 2.616 bits, so each assumption can affect the headline conclusion. The technical supplement gives the full validation record and the commands used to reproduce the smaller experiments.

Table 2: Status of the main claims.

Claim	Status
Direct system in $m - 4$ variables	Exact for every nonzero target once an initial common zero is supplied. Chevalley–Warning guarantees this zero in every seven-dimensional starting subspace, and fixing the common scale removes one variable.
Initial root lies outside the oil space	Probability bound for every fixed key over the attack's public randomness. This statement does not average over key generation.
Construction of the attack subspace	Deterministic and finite after the initial root, using linear algebra and exhaustive search in $\mathbb{P}^6(\mathbb{F}_q)$.
Probability of a usable direct root	Proved lower bound in the independent-graph random expanded-key model.
A root with full-rank derivative determines another signing key	Exact for every expanded key that passes the public rank, graph, and signing checks.
Returned signature or key is valid	Exact; every candidate is verified against the unchanged public map.
Concrete cost rows	Calculations for symbolic algorithms, not measurements from implemented circuits.
Full-parameter memory	Exceeds physical storage for the symbolic representations analyzed here.
Fixed SHAKE or AES expansion	Verification is exact for every key. The success probabilities transfer only under the stated random-function or ideal-cipher models, with explicit terms for finite output length and seed collisions.

10 Cost, memory, and parameter consequences

10.1 Work unit and scope

The QR-UOV specification expresses classical attack costs as Boolean-gate exponents. A reported cost c therefore means approximately 2^c Boolean gates. The specification charges multiplication of two b-bit field elements by

$$\mathsf{G}_{\mathrm{mul}}(b) = 2b^2 + b \tag{58}$$

Boolean gates [3]. We use the same unit. We upper-bound every extension-field operation by one multiplication using the explicit divide-and-conquer Karatsuba circuit in (55).

The main solver cost comes from multiplying the truncated polynomials that represent the moving roots. For a system with a equations in a variables,

$$C = 2^a, \qquad C_0 = 2^{a-1}(a+2),$$

the solver reconstructs through precision $2C_0$. For the concrete numbers, we assume that multiplying polynomials of length t costs

$$\mathsf{M}(t) = t \log_2 t \log_2 \log_2 t.$$

The complete power-series calculation is then charged by

$$3(a^3 + a^2)\mathsf{M}(8CC_0) \tag{59}$$

Table 3: Estimated base-two logarithms of classical Boolean-gate costs. A positive margin means that the estimated attack cost is below the category target.

	Level I		Level III		Level V	
Attack	Cost	Margin	Cost	Margin	Cost	Margin
Direct forgery, original solver	151.639	−8.639	203.507	3.493	260.389	11.611
Direct forgery, accepted-root solver	140.384	2.616	191.027	15.973	247.125	24.875
Equivalent-key recovery	154.826	−11.826	206.177	0.823	260.857	11.143
Best attack	**140.384**	**2.616**	**191.027**	**15.973**	**247.125**	**24.875**
Category target	143		207		272	

operations in the chosen working field. The direct-forgery row uses $a = m - 4$. Its cost includes the probabilities that the initial subspace avoids the oil space, that a usable root exists, that the chosen chart contains it, and that the separator identifies it. The equivalent-key row uses $a = V$ and includes the corresponding full-rank and separator probabilities. The calculation also includes the lower-order costs of constructing the equations, filtering roots, factoring polynomials, completing the graph, restoring the scale, and verifying the final output. These terms do not change the displayed decimals.

These are calculations for symbolic algorithms, not measurements from an implemented circuit. In particular, the formula for M assigns a unit constant to an asymptotic multiplication bound. The algebraic reductions, field-multiplication circuits, chosen fields, and success probabilities do not depend on this convention. A practical running time does.

10.2 The two attacks

The following table reports the estimated cost exponent and its distance from the category target for both attacks.

Both direct rows solve systems in $50, 74, 101$ variables. The second row is lower because the separator needs to distinguish only an accepted root, one qth-power equality identifies base-field roots, and coordinates are reconstructed only after a candidate passes the scalar tests. The selected extension degrees are 8, 12, and 16. The equivalent-key row uses the independent attack from Section 5. The smaller implementation executes the refined direct attack, but the full-parameter polynomial calculation still dominates both time and memory.

For context, Table 4 reconstructs the dominant storage in the published attacks. We use the most favorable standard model for each attack. For sparse linear algebra, we count three field vectors and assume that matrix entries are generated when they are needed. For PXL, we count the dense $\alpha \times \alpha$ state in the usual realization of its α^{ω} elimination step. For Just Guess, we count the quadratic coefficients and its two change-of-basis matrices using one machine word per field element. Thus the table should be read as an order-of-magnitude comparison, not as measured peak memory. The memory entries are base-two logarithms of bytes: 20 is one MiB, 40 is one TiB, and 60 is one EiB.

These estimates show that most earlier algebraic attacks also have enormous linear-algebra states. At Level I, even the matrix-free Hashimoto estimate needs about $2^{54.84}$ bytes, or 29 PiB. Rectangular MinRank needs about 3 ZiB, and the usual PXL state needs about 4.8 ZiB. Just Guess is the exception. Its depth-first search can run with tens of MiB, but its running time is much larger.

Our reduction is independent of the solver used after it. The accepted-root solver gives the smallest time estimate, but it stores all 2^a branches to a precision proportional to $a2^a$. Its main

Table 4: Time and storage for prior attacks. Each entry is ($\log_2$ time, $\log_2$ bytes). Wedge time counts field operations; all other times count gates. Storage uses a dense state for PXL and regenerated matrix entries for the sparse solvers.

Attack	Level I	Level III	Level V
PXL direct [3]	$(150.44, 72.25)$	$(210.57, 105.36)$	$(278.92, 143.62)$
Hashimoto with WXL [26]	$(158, 54.84)$	$(217, 77.92)$	$(285, 110.67)$
Just Guess [20, 21]	$(258, 23.25)$	$(399, 24.84)$	$(561, 26.11)$
Reconciliation [3]	$(164, 74.28)$	$(231, 107.10)$	$(291, 136.42)$
Rectangular MinRank [3, 18]	$(158, 71.57)$	$(219, 100.28)$	$(277, 129.13)$
Wedge [22]	$(182, 87.29)$	$(249, 120.26)$	$(328, 159.14)$

Table 5: Time and storage choices after our exact reduction. Each entry is ($\log_2$ gates, $\log_2$ bytes). The low-memory WXL rows regenerate matrix entries and count three vectors plus the reduced equations, using one 64-bit word per field element.

Solver after reduction	Level I	Level III	Level V
Accepted-root solver	$(140.38, \geq 109.41)$	$(191.03, \geq 158.51)$	$(247.13, \geq 213.40)$
PXL	$(143.64, 68.55)$	$(204.84, 101.74)$	$(272.98, 140.14)$
Fastest WXL point	$(159.22, 54.35)$	$(218.14, 79.73)$	$(285.67, 106.06)$
Low-memory WXL point	$(238.39, 21.24)$	$(367.82, 22.73)$	$(543.60, 23.18)$

array therefore has order $a4^a$ extension-field elements. This is a property of that solver, not of the $a = m - 4$ variable reduction. Table 5 applies two standard solvers to the same reduced system and exposes the resulting tradeoff.

The PXL row is only 3.26 bits slower than the accepted-root estimate at Level I and reduces the displayed storage by more than 40 bits. It still needs roughly 374 EiB, so this is not a practical implementation. The low-memory WXL row makes the opposite choice. It guesses $27, 45, 70$ variables, leaves Macaulay matrices of dimensions about $2^{16.58}$, $2^{18.09}$, and $2^{18.52}$, and uses about 2.4, 6.9, and 9.5 MiB. Its estimated times are respectively 19.61, 31.18, and 17.40 bits below the current Just Guess estimates. Thus our reduction improves both ends of the modeled frontier: it gives the smallest time estimates with the accepted-root solver, and it gives a faster route at the MiB storage scale than the previous depth-first method.

The low-memory comparison is conditional in two ways. It assumes that the reduced equations follow the semi-regular degree model used in the published WXL estimates. It also omits the ordinary integer work needed to regenerate matrix entries, just as the explicit low-memory option in the wedge analysis does. These calculations are reproducible from the published formulas with `QRUOV_memory_audit.py`; they are estimates, not full-parameter measurements.

10.3 What follows at each level

The refined direct rows lie below the three targets by 2.616, 15.973, and 24.875 bits. At Level I this is a candidate break only under the stated model. The polynomial representation cannot fit in physical memory, the fast multiplication schedule has not been implemented at full size, and the degree theorem has not been checked for every detail of the exact attack curve. The larger Level-III and Level-V margins, together with the independent equivalent-key estimates, support changing those parameter sets. The smaller implementation checks that the direct-attack steps work together. It does not make any full-parameter row practical.

10.4 Why changing only the vinegar count is not a repair

The official parameter sets sit on the direct side of the crossing between the two reductions:

Level	V	$m-4$	Key-recovery variables	Direct variables
I	52	50	52	50
III	76	74	76	74
V	102	101	102	101

Increasing V raises the key-recovery dimension but leaves the direct-forgery dimension unchanged. Decreasing V eventually makes the equivalent-key system no larger than $m-4$. Thus, with m fixed, changing only V always leaves at least one of the two systems at or below this size.

The next table applies the same conservative cost calculation to nearby parameters. It is not a complete replacement proposal. Any proposal must also recompute other attacks, failure probabilities, key sizes, and implementation costs. The table isolates the consequence proved here: raising only V leaves direct forgery unchanged, while raising only m leaves the equivalent-key system unchanged.

Table 6: Smallest (V, m) pairs with m divisible by three in the cost calculation that put both displayed attacks at the category target, or 16 bits above the target. This is not a complete parameter proposal.

	Current		Reach target		Reach target +16	
Level	V	m	V	m	V	m
I	52	54	52	54	57	60
III	76	78	78	81	87	90
V	102	105	108	111	117	120

The table enforces $m \equiv 0 \pmod 3$ and the condition $V \geq m-3$ needed by the direct reduction. Level I already reaches its category target in this model, but placing the attack cost 16 bits above the target requires increasing both dimensions. At Levels III and V, even reaching the target requires increasing both V and m. A design change that invalidates one reduction may cost less than increasing both dimensions.

10.5 Why the fastest estimate uses so much memory

The desired outputs are small. Direct forgery needs one vector in $\mathbb{F}_q^n$, and equivalent-key recovery needs a $V \times O$ graph. The solver, however, stores all $C = 2^a$ candidate branches while following them to the target equations. The parameter degree is bounded by $C_0 = 2^{a-1}(a+2)$, so rational reconstruction uses precision proportional to $a2^a$. Storing one coefficient series for every branch therefore takes $\Theta(a4^a)$ extension-field elements. One such array at the final precision already requires approximately

$$2^{49.41} \text{ EiB}, \qquad 2^{98.51} \text{ EiB}, \qquad 2^{153.40} \text{ EiB}$$

for the direct systems. The corresponding equivalent-key arrays require about $2^{53.73}$, $2^{102.68}$, and $2^{155.41}$ EiB. These quantities exceed physical storage by many orders of magnitude.

The direct reduction itself does not require this representation. PXL and WXL can solve the same $50, 74, 101$ variable cores with the time and storage choices in Table 5. In particular, the low-memory WXL choice uses ordinary MiB-scale storage by regenerating matrix entries. Its running time remains prohibitive. Thus no full-parameter forgery or equivalent-key recovery is reported. The paper treats the exact reduction, each solver estimate, working storage, and end-to-end verification as separate claims.

11 Conclusion

Changing only the number of vinegar variables cannot make both attacks harder. For every vinegar count and every nonzero verification target, one of the two attack reductions produces a square nonlinear system with at most $m - \ell - 1$ variables. For QR-UOV, $\ell = 3$, so the bound is $m - 4$. At the recommended parameter sets, the direct systems have 50, 74, and 101 variables. When the vinegar count is smaller, the equivalent-key route supplies the smaller system. This algebraic statement is exact and does not depend on the cost of a particular equation solver.

We also estimate the work needed to solve these systems. With the adapted accepted-root solver, the estimates lie below all three category targets and the Level-I margin is 2.616 bits. Applying PXL to the same cores gives a second set of estimates, and low-memory WXL gives a route with only MiB-scale storage. These different points all come from the same exact reduction. The structural result is therefore not tied to one unusually memory-intensive solver.

The smaller implementation produces forged signatures from expanded public keys and verifies them with public data. It confirms the algebraic steps, but not the full-parameter cost. The concrete estimates still rely on the stated degree theorem, polynomial multiplication cost, semi-regularity where WXL or PXL is used, and the random expanded-key model. The fastest rows require more memory than any physical system can provide. The MiB-scale rows have prohibitive running time. The firm design conclusion is independent of these tradeoffs: a parameter revision must increase the output dimension as well as the vinegar dimension, or change the construction so that one of the two reductions no longer applies.

References

[1] H. Furue, Y. Ikematsu, Y. Kiyomura, and T. Takagi. A new variant of Unbalanced Oil and Vinegar using quotient ring: QR-UOV. Cryptology ePrint Archive, Paper 2020/1243, 2020.

[2] P. Li, H. Guo, and J. Ding. Unified approach to UOV-like multivariate signature schemes. Cryptology ePrint Archive, Paper 2025/1778, 2025.

[3] H. Furue, Y. Ikematsu, F. Hoshino, T. Takagi, H. Kosuge, K. Yamakoshi, R. Akiyama, S. Nakamura, S. Orihara, and K. Kinjo. *QR-UOV Specification Document (Round 2 submission)*. NIST Additional Digital Signatures Project, 5 February 2025.

[4] A. Kipnis, J. Patarin, and L. Goubin. Unbalanced Oil and Vinegar signature schemes. In *EUROCRYPT 1999*, pp. 206–222, 1999.

[5] A. Kipnis and A. Shamir. Cryptanalysis of the Oil and Vinegar signature scheme. In *CRYPTO 1998*, pp. 257–266, 1998.

[6] A. Braeken, C. Wolf, and B. Preneel. A study of the security of Unbalanced Oil and Vinegar signature schemes. Cryptology ePrint Archive, Paper 2004/222, 2004.

[7] W. Beullens and B. Preneel. Field lifting for smaller UOV public keys. In *INDOCRYPT 2017*, pp. 227–246, 2017.

[8] A. Szepieniec and B. Preneel. Block-anti-circulant Unbalanced Oil and Vinegar. In *SAC 2019*, pp. 574–588, 2019.

[9] H. Furue, K. Kinjo, Y. Ikematsu, Y. Wang, and T. Takagi. A structural attack on block-anti-circulant UOV at SAC 2019. In *PQCrypto 2020*, pp. 323–339, 2020.

[10] J. Ding, B.-Y. Yang, C.-O. Chen, M.-S. Chen, and C.-M. Cheng. New differential-algebraic attacks and reparametrization of Rainbow. In *ACNS 2008*, pp. 242–257, 2008.

[11] National Institute of Standards and Technology. Nine candidates advance to the third round of the Additional Digital Signatures for the PQC standardization process. NIST CSRC, 14 May 2026.

[12] G. Alagic et al. Status report on the second round of the Additional Digital Signature Schemes for the NIST Post-Quantum Cryptography Standardization Process. NIST IR 8610, May 2026.

[13] J. van der Hoeven and G. Lecerf. On the complexity exponent of polynomial system solving. *Foundations of Computational Mathematics*, 21(1):1–57, 2021.

[14] M. Giusti, G. Lecerf, and B. Salvy. A Gröbner free alternative for polynomial system solving. *Journal of Complexity*, 17(1):154–211, 2001. doi:10.1006/jcom.2000.0571.

[15] N. Giménez, J. Heintz, G. Matera, L. M. Pardo, M. Pérez, and M. Privitelli. A Kronecker algorithm for locally closed sets over a perfect field. arXiv:2512.14888v2, 5 June 2026.

[16] W. Beullens. Improved cryptanalysis of UOV and Rainbow. In *EUROCRYPT 2021, Part I*, pp. 348–373, 2021.

[17] H. Furue and Y. Ikematsu. A new security analysis against MAYO and QR-UOV using rectangular MinRank attack. In *IWSEC 2023*, pp. 101–116, 2023.

[18] T. Suzuki, H. Furue, T. Ito, S. Nakamura, and S. Uchiyama. An extended rectangular MinRank attack against UOV and its variants. In *IWSEC 2025*, LNCS 16208, pp. 111–130, Springer, 2026.

[19] H. Furue and Y. Ikematsu. Key recovery attacks on UOV using p^ℓ-truncated polynomial rings. Cryptology ePrint Archive, Paper 2026/298, 2026.

[20] A. May, M. Ostuzzi, and H. Ressler. Just Guess: Improved (quantum) algorithm for the underdetermined MQ problem. In *EUROCRYPT 2026, Part IV*, LNCS 16544, pp. 334–362, 2026.

[21] H. Asanuma, Y. Chen, H. Furue, K. Sakata, and T. Takagi. Improved complexity estimates for underdetermined MQ systems via generalized variable partitioning. Cryptology ePrint Archive, Paper 2026/1054, 2026.

[22] L. Ran. Wedges, oil, and vinegar: An analysis of UOV in the exterior algebra. Cryptology ePrint Archive, Paper 2025/1143, 2025.

[23] P. Pébereau. One vector to rule them all: Key recovery from one vector in UOV schemes. In *PQCrypto 2024*, pp. 92–108, 2024; ePrint 2023/1131.

[24] M. Safey El Din and E. Schost. Bit complexity for multi-homogeneous polynomial system solving: Application to polynomial minimization. arXiv:1605.07433v2, 2017; manuscript revised 6 November 2018.

[25] E. Schost. Computing parametric geometric resolutions. *Applicable Algebra in Engineering, Communication and Computing*, 13(5), 2002. doi:10.1007/s00200-002-0109-x.

[26] S. Orihara, on behalf of the QR-UOV team. Round 2 (Additional Signatures) official comment: QR-UOV. NIST PQC Forum, 20 January 2026.

[27] Y. Hashimoto. An improvement of algorithms to solve under-defined systems of multivariate quadratic equations. *JSIAM Letters*, 15:53–56, 2023. doi:10.14495/jsiaml.15.53.

[28] E. Thomae and C. A. Wolf. Solving underdetermined systems of multivariate quadratic equations revisited. In *PKC 2012*, LNCS 7293, pp. 156–171, 2012. doi:10.1007/978-3-642-30057-8_10.

[29] H. Furue, S. Nakamura, and T. Takagi. Improving Thomae–Wolf algorithm for solving underdetermined multivariate quadratic polynomial problem. In *PQCrypto 2021*, LNCS 12841, pp. 65–78, 2021. doi:10.1007/978-3-030-81293-5_4.

[30] C. Chevalley. Démonstration d'une hypothèse de M. Artin. *Abhandlungen aus dem Mathematischen Seminar der Universität Hamburg*, 11:73–75, 1935.

[31] E. Warning. Bemerkung zur vorstehenden Arbeit von Herrn Chevalley. *Abhandlungen aus dem Mathematischen Seminar der Universität Hamburg*, 11:76–83, 1935.

[32] Y. Jin, Y. Pan, X. He, B. Gong, and J. Ding. Security analysis on UOV families with odd characteristics: Using symmetric algebra. Cryptology ePrint Archive, Paper 2025/1137; in *PKC 2026, Part I*, LNCS 16551, pp. 428–454, 2026.